

Shanmukha Pratheek Annadanam

AI / Machine Learning Engineer

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Work Experience

Iter8 (Self-funded AI Startup) - Cursor for Figma (React, TypeScript, Node.js, Express, Anthropic SDK, SSE) Remote
Founding Engineer Dec 2025 - Apr 2026

- Developed and deployed a production LLM-based application end to end, owning model integration, prompt engineering, backend services, deployment, and production monitoring as the sole engineer
- Applied prompt engineering and structured output design so model responses were consistent and machine-readable, then optimized cost by roughly 85% through prompt caching after profiling where compute was actually spent
- Optimized application performance under real load with a streaming architecture over Server-Sent Events and incremental diff computation, holding sub-second response times
- Ensured output correctness through a bidirectional transformation layer validated at every stage, achieving lossless accuracy across 85+ structured properties before downstream consumption
- Troubleshoot model, application, and production issues across service and integration layers, including reading third-party SDK internals to resolve undocumented behavior

Brane Group Hyderabad, Telangana
Software QA Intern Sep 2023 - Jun 2024

- Built end-to-end automated test suites for critical user flows in an enterprise application, reducing regression testing time by 30%, and triaged failures to root cause across application and integration layers

AI / ML Project Experience

Multi-Agent Research Assistant (Python, FastAPI, Qdrant, Sentence-Transformers, Anthropic Claude + Gemini, Tavily, React, Docker, AWS ECS/ECR, Langfuse, RAGAS)

- Built a RAG pipeline end to end: sentence-transformer embeddings, indexing and semantic retrieval from a Qdrant vector database, and a synthesis stage that deduplicates and reconciles conflicting sources before returning a ranked answer
- Designed AI agents and automated workflows: three specialized agents (Retriever, Web Researcher, Synthesizer) run under an async Python orchestrator handling planning, tool calling, shared state, retries, and automatic fallback between LLM providers
- Built the backend services and REST APIs with FastAPI, streaming results to a React client, and integrated multiple LLM providers (Claude primary, Gemini fallback) plus external search APIs
- Implemented model evaluation and monitoring: automated LLM evaluation with RAGAS (0.88 faithfulness, 0.87 answer relevancy) gating releases, and Langfuse tracing of per-step latency, token cost, and failure rates
- Applied MLOps practices for production deployment: Docker containerization, AWS ECS Fargate with images in ECR, and CloudWatch monitoring on Linux containers

Customer Churn Intelligence Platform (Python, SQL, Pandas, scikit-learn, XGBoost, PyTorch, SHAP, FastAPI, Docker, Streamlit)

- Prepared, cleaned, transformed, and analyzed a 7,043-record dataset: SQL extraction and aggregation, outlier removal, 7 engineered features, and stratified train/validation/test splitting
- Trained, evaluated, and optimized models across 5 families with 5-fold cross-validation (0.8468 ROC-AUC, 78.3% recall), including a 3-layer PyTorch neural network with batch normalization, dropout, and class-weighted loss, plus SHAP explainability
- Deployed the model behind a FastAPI REST service (7 endpoints) with Docker and a monitoring dashboard for prediction quality

Education

University at Albany, SUNY | GPA: 3.8 Albany, New York
Master of Science in Data Science May 2026

Keshav Memorial Institute of Technology Hyderabad, Telangana
Bachelor of Technology in Computer Science (AI and Machine Learning) May 2024

Technical Skills

Languages	Python (strong), SQL, TypeScript, JavaScript, C++
Generative AI / LLMs	Anthropic Claude, Google Gemini, Prompt Engineering, LLM Evaluation, Structured Outputs, Tool Calling, AI Agents & Intelligent Automation, Multi-Agent Orchestration, AI-Powered Workflows, Prompt Caching
RAG & Vector	Retrieval-Augmented Generation, Vector Databases (Qdrant), Embeddings (Sentence-Transformers), Semantic & Similarity Search, Document Processing, NLP
ML & Deep Learning	PyTorch, TensorFlow, scikit-learn, XGBoost, Neural Networks, Feature Engineering, Cross-Validation, Model Evaluation & Optimization, SHAP
MLOps & Monitoring	Model Monitoring, LLM Evaluation Harnesses (RAGAS), Tracing & Observability (Langfuse), Performance & Cost Optimization, CI Workflows, Automated Testing
APIs & Backend	FastAPI, REST API Design, Node.js, Express, SSE Streaming, Async Orchestration, Webhooks
Cloud & DevOps	AWS (ECS, ECR, Fargate, CloudWatch), Docker, Containerization, Linux, Git & GitHub
Data	SQL & Relational Database Fundamentals, Pandas, NumPy, Data Cleaning & Transformation, ETL Pipelines, Tableau, Power BI, Streamlit

Certifications: AI Fluency: Framework & Foundations, Anthropic Education (Jul 2026) | Claude 101, Anthropic Education (Jul 2026)